

IPv6 Enabling Peer-to-Peer IMS Services



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Agenda

- IP multimedia services
- IMS and IPv6
- Example of peer-to-peer IP connectivity
- Technology and application trends
- Multi-access IMS
- Conclusions

IP multimedia services

The four key changes



Richer person-to-person communication

- IP will increase usage through reachability of users, seamless usage of different communication types and easy-to-use group communications.

Richer interaction between media streams

- Easy integration and interworking of different IP-based services.

Service mobility

- Consistent services over various different access networks increase usage and reduce churn.

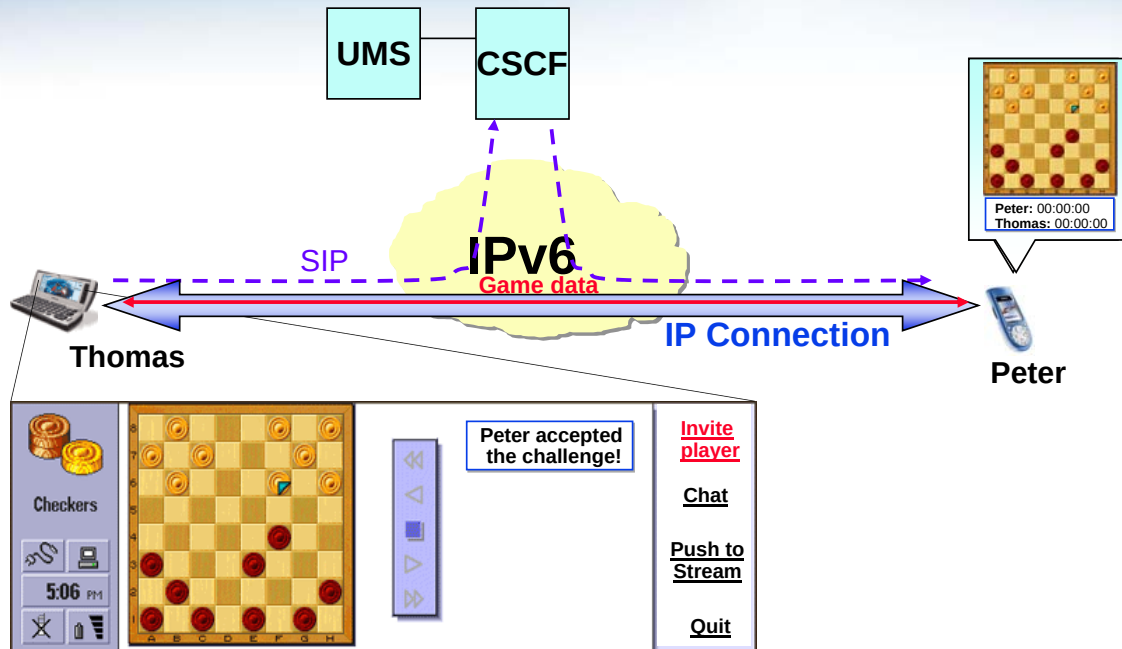
Easier service creation and integration

- Well defined open APIs and programming languages enable service creation by 3rd party developers.

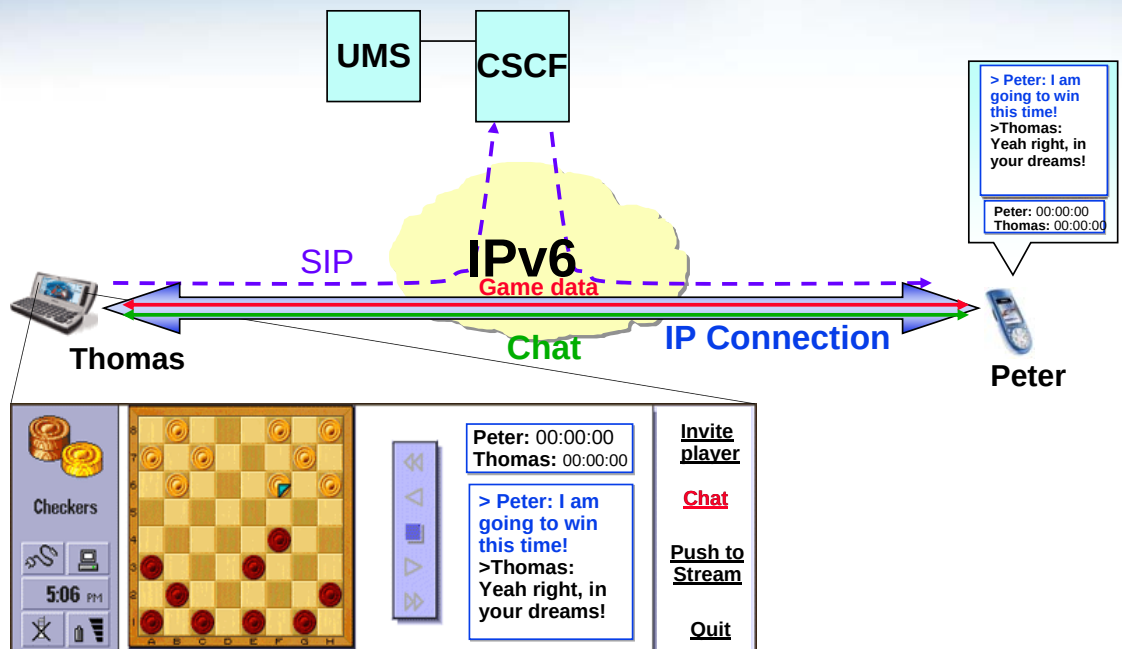
Global SIP/IMS deployment needs IPv6

- Introduction of SIP-based peer-to-peer services is an important step after current client-server based services.
- IP Multimedia Subsystem (IMS) is a service infrastructure based on the use of Session Initiation Protocol (SIP).
 - 3GPP Release 5 and 6 specifications
 - 3GPP2 specifications
- In order to make peer-to-peer services work between different operators' networks, IPv6 is needed - peer-to-peer services work well only with public IP addresses.
 - Small scale IMS deployment / piloting can be started with IPv4.
 - IPv6 is vital for wider scale, global IMS deployment.

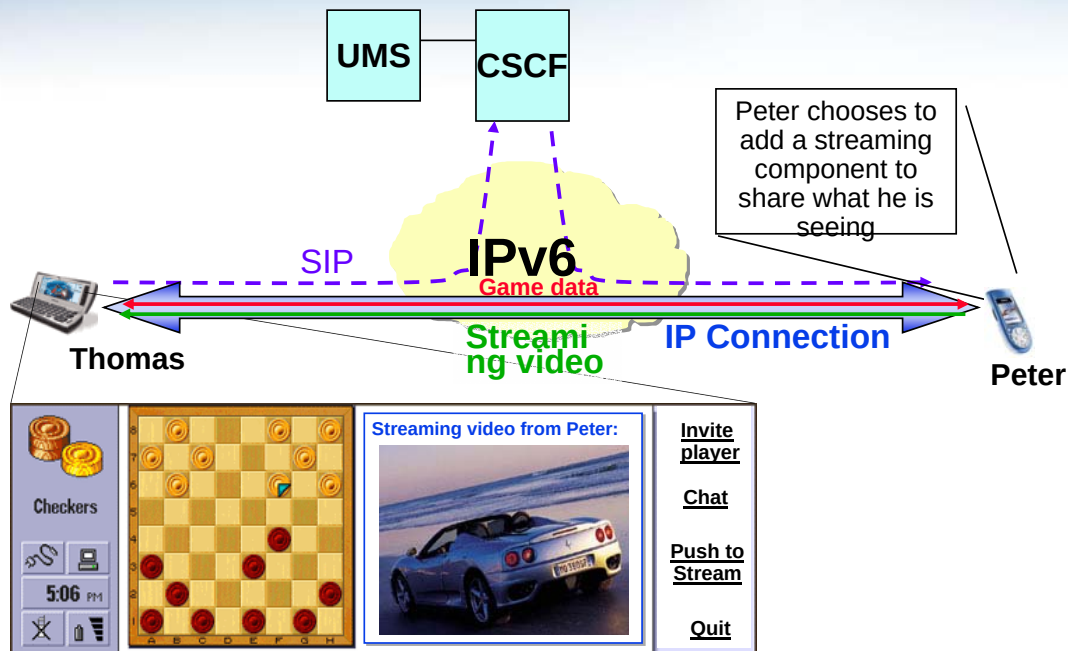
Example of peer-to-peer IP connectivity



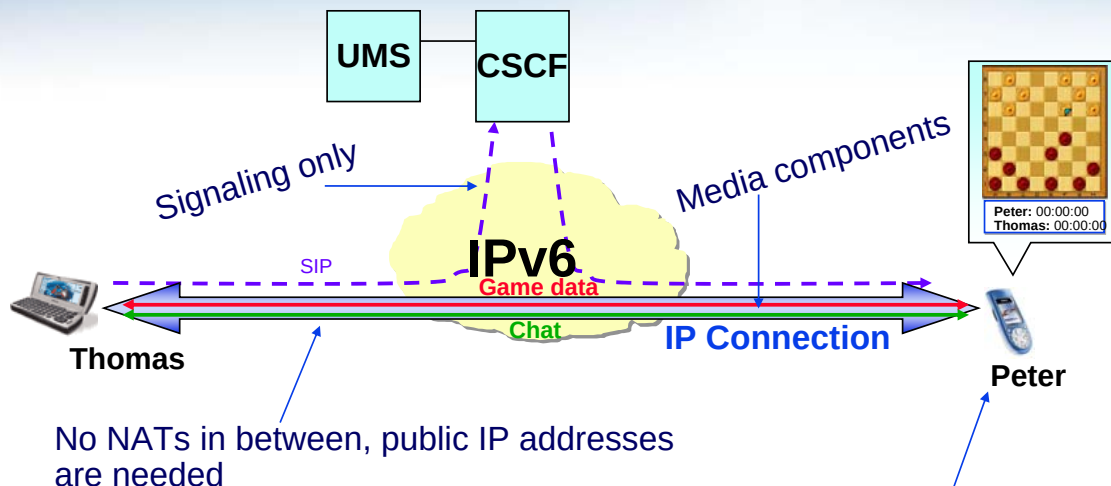
Example of peer-to-peer IP connectivity



Example of peer-to-peer IP connectivity



Future mobile services = "serverless media"



The SIP/IMS user plane is peer-to-peer in nature - SIP/IMS sessions between mobiles in different Private IPv4 address spaces become highly complicated. This is why public IP addresses are required. The only future proof solution is provided by IPv6.

Example services: gaming, chat, streaming, Voice/video over IP, etc.

Standardized technology enablers for new mobile services are here today



IPv6 **Color displays** **SIP**

MMS **Imaging and camera integration** **XHTML and TCP/IP**

Video **Multimedia Streaming** **Multimode** **GPRS** **EDGE** **WCDMA** **CDMA2000** **Bluetooth** **WLAN**

Symbian **Positioning** **MIDI** **Presence** **Java** **DRM**

Options **Back** **Options** **Exit**

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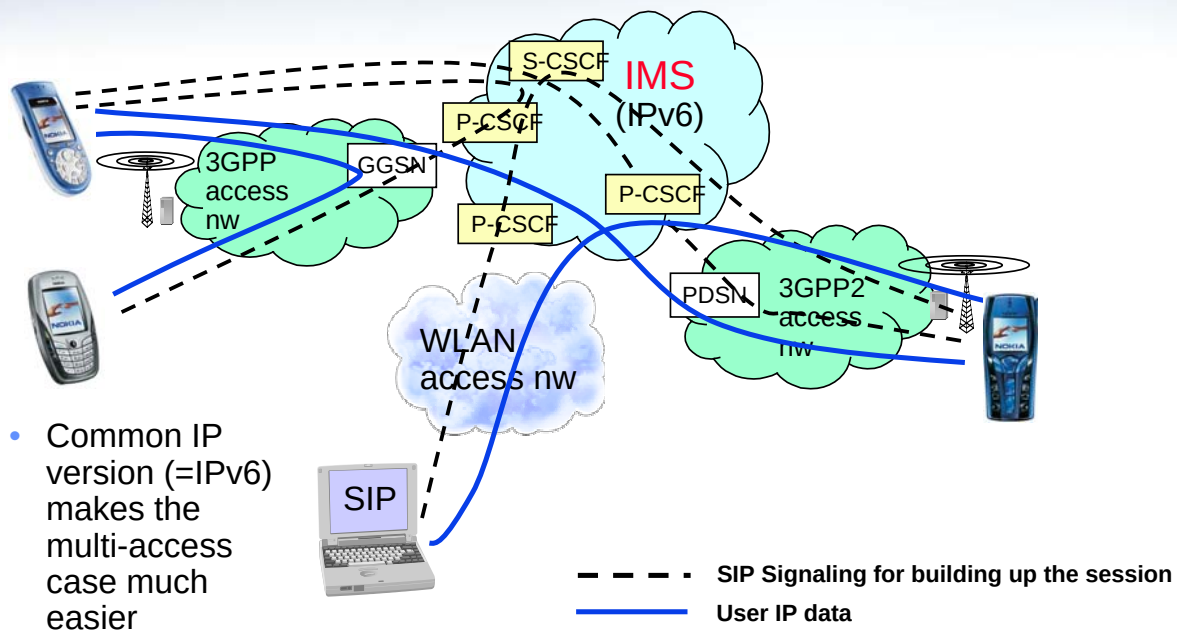
Technology and application trends



- 2G radio interface (GSM / EDGE)
- IPv4
- Client-server connectivity
- SMS text messaging, WAP browsing, MMS multimedia messaging

- 2G and 3G radio interfaces (WCDMA / CDMA2000)
- IPv4/IPv6 dual stack
- Peer-to-peer connectivity
- Richer, IP-based Applications
 - HTTP/TCP/IP browsing
 - Presence
 - Instant Messaging
 - Multimedia streaming
 - Gaming
 - Voice and video telephony
 - Sharing
 - Etc.

Multi-access IMS



Conclusions

- Introduction of SIP-based peer-to-peer services is an important step after current client-server based services.
- IPv6 enables global IMS-based peer-to-peer services.
- IMS services will in the first phase be accessed via 3GPP and 3GPP2 radio access networks. WLAN will follow.
- Communications between cellular terminals and Internet clients (WLAN and fixed networks) enables successful business.

The image features a large, stylized blue 'N' on the left side, which is part of the Nokia logo. The background is white with a subtle blue wave-like pattern. The word 'NOKIA' is written in a bold, blue, sans-serif font. Below it, the tagline 'CONNECTING PEOPLE' is written in a smaller, blue, serif font.

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CONNECTING PEOPLE